

InnovAero Competition 2026

VIP Aircraft Cabin Acoustic Comfort Enhancement

Academic report structure prepared for technical review

Team	SlowWaveAero
Authors	Yossef Al Buni; Leander Meggers
Affiliation	Technische Universität Hamburg (TUHH) and NORDAKADEMIE Hochschule der Wirtschaft
Client context	Lufthansa Technik / InnovAero Competition 2026
Date	May 30, 2026

Abstract

This report consolidates the literature and design logic most relevant to the InnovAero task of improving acoustic comfort in a VIP aircraft cabin. The focus is on methods that are technically defensible and implementation-oriented: active noise control, adaptive filtering, virtual sensing, cabin-specific acoustic constraints, and hybrid passive-active concepts. The resulting structure is intended to support an academically rigorous submission that can be extended with competition-specific calculations, design assumptions, and verification data.

Contents

1	Introduction	1
1.1	Motivation	1
1.2	Report Purpose	1
1.3	Document Structure	1
2	Task Definition and Requirements	2
2.1	Competition Task	2
2.2	Key Requirements	2
2.3	Engineering Constraints	2
2.4	Success Criteria	2
3	Related Work	3
3.1	Problem Context	3
3.2	Active Noise Control Fundamentals	3
3.3	Aircraft-Cabin-Specific Constraints	3
3.4	Localized and Seat-Level Control	4
3.5	Passive, Active, and Hybrid Concepts	4
3.6	Data-Driven and Computational Approaches	4
3.7	Synthesis and Research Gap	5
3.8	Implications for the Competition Deliverable	5
3.9	Conclusion	5
4	System Concept	6
4.1	Concept Overview	6
4.2	Architecture	6
4.3	Why This Architecture Fits the Task	6
4.4	Design Principles	6
5	Methodology	7
5.1	Analysis Approach	7
5.2	Evaluation Metrics	7
5.3	Validation Plan	7
5.4	Expected Report Outputs	7
6	Conclusion and Outlook	8
6.1	Conclusion	8
6.2	Outlook	8

List of Tables

3.1	High-level comparison of cabin noise-control strategies	4
-----	---	---

1 Introduction

1.1 Motivation

Passenger comfort is a differentiating factor in premium aircraft cabins, especially in the VIP segment where the interior is expected to support work, rest, and quiet conversation. In this context, cabin noise is not only an engineering nuisance but also a design constraint that influences perceived quality, fatigue, and overall product value. The InnovAero task therefore requires a solution that is technically credible, compact, and compatible with aviation integration constraints.

1.2 Report Purpose

This report is structured to support a professional competition submission for Lufthansa Technik. It first defines the task and the relevant constraints, then reviews the literature, and finally derives a concept and a validation approach. The goal is not to present a finished certification package, but to establish a clear engineering argument that can be extended with calculations, simulations, and prototype data.

1.3 Document Structure

The following chapters provide a logical flow from context to solution.

- Chapter 2 summarizes the competition task and the engineering requirements.
- Chapter 3 reviews the most relevant literature on cabin noise control.
- Chapter 4 proposes a system-level concept for the VIP cabin use case.
- Chapter 5 outlines a methodology for analysis, simulation, and evaluation.
- Chapter 6 concludes the report and identifies the next development steps.

2 Task Definition and Requirements

2.1 Competition Task

The underlying task is to improve acoustic comfort in a VIP aircraft cabin using methods that are suitable for an engineering proposal delivered to Lufthansa Technik. The problem is inherently multidisciplinary because it combines acoustics, signal processing, structural integration, usability, and maintainability. A strong concept must therefore be evaluated not only by theoretical cancellation performance, but also by its feasibility as an aircraft interior solution.

2.2 Key Requirements

From an aerospace systems perspective, the most important requirements are low mass, low integration complexity, limited maintenance burden, robust operation under cabin variability, and a clear safety boundary around the active control functions. For a premium cabin, the design must also preserve the visual and spatial quality of the interior.

2.3 Engineering Constraints

The most relevant constraints are summarized below.

1. The solution should target the dominant low-frequency noise components where active methods are most effective.
2. The concept should remain localized enough to be installed around a seat group or premium zone rather than requiring full-cabin retrofit.
3. The architecture should tolerate model mismatch and operational variation, especially in the secondary path.
4. The design should allow passive trim to carry part of the load so that the active layer can remain compact.
5. The evaluation method should include both physical performance and passenger-facing acoustic quality.

2.4 Success Criteria

A successful submission should show that the chosen concept reduces the perceptually relevant noise components, can be integrated into a realistic cabin architecture, and is defensible against a review by aviation engineers. In practice, this means the report should explicitly connect the literature, the selected control strategy, and the expected implementation path.

3 Related Work

3.0.1 Motivation and scope

The acoustic-comfort problem in a VIP aircraft cabin can be formulated not only as a materials and insulation problem, but also as a signal-processing problem. The passenger perceives a disturbance signal that is generated by one or several primary sources, transmitted through a structural-acoustic path, modified by the cabin geometry and interior panels, and finally received at the passenger's ears. In a simplified discrete-time formulation, the undesired sound pressure at an error microphone or at the passenger's ear can be written as

$$d(n) = (p * x)(n) + v(n), \quad (3.1)$$

where $x(n)$ is a measurable reference signal correlated with the engine or structural excitation, $p(n)$ denotes the primary acoustic or vibro-acoustic transmission path, $*$ denotes convolution, and $v(n)$ collects uncorrelated background components and measurement noise.

A purely passive solution reduces $d(n)$ by modifying the transmission path $p(n)$ through absorption, damping, isolation, and transmission-loss improvements. An active solution adds a secondary sound or structural response $y(n)$ such that the residual error

$$e(n) = d(n) + (s * y)(n) \quad (3.2)$$

becomes small at the target location. Here, $s(n)$ is the secondary path from the secondary actuator, e.g., loudspeaker, trim-panel actuator, or inertial exciter, to the error sensor. The standard objective is the minimisation of the mean-square acoustic error,

$$J = \mathbb{E}\{e^2(n)\}, \quad (3.3)$$

which directly connects aircraft-cabin noise control with adaptive filtering, system identification, channel equalisation, and multi-input multi-output (MIMO) digital communication.

This section reviews the most relevant literature for treating aircraft-cabin noise reduction as a mathematical signal-processing problem. The literature can be grouped into five areas: general active noise control (ANC), aircraft-cabin ANC, local headrest-based quiet-zone generation, active trim-panel and sidewall systems, and hybrid passive-active solutions.

3.0.2 General active noise control and adaptive signal processing

Kuo and Morgan provide one of the most influential tutorial reviews of active noise control and establish the signal-processing foundation of ANC systems [5]. Their work formulates ANC as the generation of a cancelling anti-noise wave by secondary sources. In digital implementations, the controller is normally represented by an adaptive finite impulse response (FIR) filter

$$y(n) = \mathbf{w}^T(n)\mathbf{x}(n), \quad (3.4)$$

where $\mathbf{x}(n) = [x(n), x(n-1), \dots, x(n-L+1)]^T$ is the reference-signal vector and $\mathbf{w}(n)$ contains the adaptive filter coefficients. A major contribution of this work is the detailed explanation of the filtered-x least-mean-square (FxLMS) algorithm, which modifies the classical LMS update because the anti-noise signal passes through the secondary acoustic path $S(z)$ before it reaches the error microphone. The coefficient update is commonly written as

$$\mathbf{w}(n+1) = \mathbf{w}(n) - \mu e(n)\mathbf{x}_f(n), \quad (3.5)$$

where μ is the adaptation step size and $\mathbf{x}_f(n)$ is the reference vector filtered by an estimate $\hat{S}(z)$ of the secondary path. For the present VIP-cabin task, this means that the loudspeaker–cabin–ear transfer function cannot be ignored. Any active cancellation system integrated into a seat, sidewall, or ceiling panel must estimate or identify the secondary path before stable adaptation is possible.

Elliott and Nelson review active noise control from both the physical-acoustics and signal-processing perspectives [4]. Their main argument is highly relevant for aircraft interiors: active control is particularly attractive at low frequencies, where passive absorption and insulation become bulky, heavy, or inefficient. They also explain that ANC is limited by wavelength, sensor placement, actuator placement, causality, and the spatial extent of the controlled sound field. For a VIP cabin, this implies that an active system should preferably target low-frequency tonal and narrowband components, while passive materials should be used for mid- and high-frequency broadband noise. Their review is also important because it frames ANC within control theory and signal processing, allowing the cabin-acoustic problem to be treated mathematically as feedback/feedforward control of a physical acoustic field.

Nelson and Elliott’s monograph further develops the physical and mathematical foundations of active sound control [6]. A key insight is that cancellation at a single point does not automatically imply global reduction of sound energy in a three-dimensional enclosure. Instead, the achievable quiet zone depends on frequency, wavelength, sensor geometry, and source–receiver configuration. This is important for the InnovAero cabin task because the passenger can be seated or reclined. If the ear position changes significantly, a controller designed for one point may not produce equivalent attenuation at another point. A robust design should therefore either control multiple error positions or create a local quiet zone around the head.

3.0.3 Aircraft cabin active noise control

Aircraft cabins are complex vibro-acoustic systems. Disturbances can enter through airborne transmission, structure-borne transmission, and panel radiation. Johansson’s work on active noise control in aircraft highlights that aircraft interior noise is especially challenging because large cabins require multiple sensors and actuators, and because acoustic paths vary with operating conditions [7]. The work supports a central design conclusion for VIP cabins: ANC should not be considered a stand-alone replacement for passive insulation, but as a complementary technology for frequency ranges and transmission paths where passive measures are inefficient.

Misol presents the design of an ANC system for aircraft interior noise and focuses on the practical problem of actuator and sensor placement in the aircraft cabin [8]. The work uses a frequency-domain FxLMS approach and emphasises that ANC can effectively reduce tonal or multi-tonal aircraft interior noise up to a few hundred hertz. The placement of actuators, such as inertial exciters, and error sensors, such as microphones, is treated as a main design task. This is directly transferable to the VIP cabin: the team should not only propose a controller algorithm, but also justify where reference sensors, error microphones, and secondary sources can be installed. Possible positions include the seat headrest, sidewall panels, ceiling panels, or acoustic trim modules.

Kletschkowski provides an overview of active control of low-frequency aircraft cabin noise caused by internal and external sources, including engines and turbulent flow [9]. The paper is valuable for the conceptual argument that the low-frequency range is the most promising target for active treatments. In a long-haul VIP cabin, passenger comfort during rest or sleep is strongly affected by low-frequency tonal components and modulation effects. These components can remain perceptible even when high-frequency cabin noise is already reduced by premium interior panels. Therefore, a hybrid passive–active concept can be justified not only by acoustics, but also by psychoacoustic comfort.

3.0.4 Local quiet-zone generation near aircraft seats

For VIP cabins, local active noise control near the passenger's head is particularly attractive. Instead of controlling the entire cabin volume, the controller aims to reduce sound pressure at or near the ears. Dimino et al. experimentally assessed an ANC system for aircraft cabin seat headrests using two loudspeakers placed on both sides of the passenger's head [10]. Their system aims to create a local quiet zone around the ears and uses a two-input/two-output filtered-x LMS algorithm. This architecture is highly relevant for VIP seating because the seat already defines the passenger's expected head location in the seated position. In a reclined position, the same idea can be extended by adding secondary loudspeakers or microphones to the headrest, shell, sidewall, or sleeping surface.

Mathematically, a local headrest ANC system can be modelled as a MIMO adaptive filtering problem. If M secondary sources and K error microphones are used, the residual error vector is

$$\mathbf{e}(n) = \mathbf{d}(n) + \sum_{m=1}^M \mathbf{s}_m(n) * y_m(n), \quad (3.6)$$

where $\mathbf{e}(n) \in \mathbb{R}^K$ contains the error signals at the microphones and $\mathbf{s}_m(n)$ contains the secondary paths from source m to all error locations. The multi-channel cost function becomes

$$J = \mathbb{E}\{\mathbf{e}^T(n) \mathbf{Q} \mathbf{e}(n)\}, \quad (3.7)$$

where $\mathbf{Q}$ can weight the ears or microphone positions differently. This formulation is useful for the competition because it allows the team to explicitly prioritise comfort at the passenger's ears, rather than globally reducing sound pressure everywhere in the compartment.

Mylonas et al. investigate low-computational-complexity local ANC for adjacent aircraft seats [11]. Their work is important because multi-channel FxLMS systems can become computationally expensive when several seats, microphones, and loudspeakers are involved. They propose a mixed-error FxLMS strategy that reduces complexity while maintaining relevant attenuation performance in a headrest application. For a VIP cabin, this supports a practical architecture in which the active system is local, modular, and computationally efficient. Such a system could be embedded into a premium seat module without requiring a cabin-wide array of loudspeakers.

A later virtual-sensing approach by Mylonas et al. addresses a practical constraint of local ANC: the ideal error microphone location is often at the passenger's ear, but placing a physical microphone directly there is intrusive or impossible [12]. Virtual sensing estimates the sound pressure at a target location from microphones placed at more acceptable positions, for example behind or near the headrest. This is highly relevant for VIP-cabin design because the passenger should not be surrounded by visible sensors. In mathematical terms, virtual sensing introduces an additional estimation model,

$$\hat{e}_{\text{ear}}(n) = \mathbf{g}^T \mathbf{z}(n), \quad (3.8)$$

where $\mathbf{z}(n)$ denotes signals from remote monitoring microphones and $\mathbf{g}$ is an estimated mapping from monitoring locations to the virtual ear position. This allows the controller to minimise the estimated ear error rather than the microphone error itself.

3.0.5 Active trim panels, sidewall systems, and structural-acoustic control

A second important research direction is the use of active trim panels or active sidewall modules. Instead of emitting anti-noise directly from loudspeakers near the passenger, these systems actively control vibrating cabin panels so that less sound is radiated into the cabin. Misol and Algermissen discuss active sidewall panels with virtual microphones for aircraft interior noise reduction [13]. Their research shows that lining-integrated active modules can be designed to remain geometrically compatible with passive lining components. This is attractive for an

aircraft cabin because integration, maintainability, and certification constraints strongly limit the acceptable hardware footprint.

Wang et al. introduce deep-learning-based active trim panels for aircraft noise control [14]. Their approach combines active noise control with machine learning, using selective fixed-filter ANC for multi-tonal noise whose frequencies vary with engine rotational speed and other operating conditions. The relevance for the VIP-cabin task is twofold. First, aircraft interior noise can contain tonal or multi-tonal components that shift with operating condition. Second, a fixed-filter bank or classifier-assisted selection mechanism can reduce computational complexity and improve robustness compared with a continuously adapting controller. From a digital signal-processing perspective, the idea resembles adaptive modulation or channel-state-dependent filter selection in communication systems: instead of using one universal controller, the system selects a suitable controller according to the estimated disturbance state.

Liu et al. study passive and active interior noise control of box structures using the structural intensity method [3]. Although the geometry is simplified, box structures are useful analogies for cabins of vehicles and aircraft. The authors use finite-element modelling and structural intensity to identify dominant vibrating panels and interior noise sources. For the InnovAero task, this work is relevant because the competition explicitly asks for the identification of dominant sound-transmission paths. The structural-intensity approach provides a systematic way to decide which wall, ceiling, floor, or side panel should receive additional passive damping or active treatment. In a simplified model, the acoustic cabin can be represented by transfer functions from structural panels to passenger-ear sound pressure,

$$P_{\text{ear}}(\omega) = \sum_{i=1}^N H_i(\omega) V_i(\omega), \quad (3.9)$$

where $V_i(\omega)$ is the vibration velocity of panel i and $H_i(\omega)$ is the acoustic transfer function from that panel to the ear. Dominant-path identification then means finding panels or paths with the largest contribution to $|P_{\text{ear}}(\omega)|$.

3.0.6 Hybrid passive–active solutions

The literature consistently suggests that the most realistic aircraft-cabin solution is hybrid. Passive materials are robust, certifiable, fail-safe, and effective at mid and high frequencies. However, low-frequency engine and structure-borne noise is difficult to attenuate passively because the required mass and thickness become problematic for aircraft. Active methods are most attractive in precisely this low-frequency range, especially for tonal or narrowband disturbances. Parker’s aerospace white paper discusses passive, active, and hybrid solutions for aircraft interior noise and argues that hybrid treatment can exploit the strengths of both approaches [2]. While this source is industrial rather than purely academic, it is useful for motivating an engineering concept that combines certified passive panels with active low-frequency cancellation.

For the VIP-cabin task, a suitable hybrid architecture can be summarised as follows:

1. Identify the dominant sound-transmission path from the engine and fuselage structure to the VIP compartment.
2. Apply lightweight passive acoustic materials to the full compartment envelope to reduce broadband transmission and improve mid/high-frequency comfort.
3. Add local ANC at the headrest or sleeping area to reduce low-frequency components at the ears.
4. Optionally integrate active trim or sidewall panels if the dominant path is panel radiation from the window-side fuselage or cabin lining.
5. Use virtual sensing to estimate ear-level pressure without placing microphones directly at the passenger’s ears.

3.0.7 Connection to digital communication theory

The mathematical relationship between cabin ANC and digital communication is strong. In digital communication, a transmitted signal is distorted by a physical channel and corrupted by noise. The receiver applies channel estimation, equalisation, filtering, and sometimes adaptive interference cancellation. In aircraft-cabin ANC, the disturbance signal is transmitted through an acoustic or structural-acoustic channel and then reduced by adaptive filtering. The analogy is shown in ??.

Table 3.1: Analogy between digital communication and aircraft-cabin active noise control.

Digital communication concept	Aircraft-cabin noise-control interpretation
Physical channel $H(z)$	Acoustic or vibro-acoustic path from engine/fuselage to passenger ear
Channel estimation	Identification of primary and secondary acoustic paths $P(z)$ and $S(z)$
Equalisation	Compensation of path dynamics through adaptive filters
Interference cancellation	Generation of anti-noise to reduce correlated disturbance components
MIMO communication	Multiple microphones and loudspeakers/actuators in a cabin
Adaptive receiver	FxLMS, frequency-domain FxLMS, or selective fixed-filter controller
State-dependent transmission	Controller selection depending on engine speed, flight condition, or tonal frequencies

This analogy justifies the use of signal-processing terminology and methods in the competition report. The acoustic cabin can be modelled as a channel with transfer function $H(f)$, and passive materials modify the magnitude and phase of this channel. Active noise control then acts as an adaptive interference-cancellation system. If the engine signal is sufficiently correlated with the cabin noise, a feedforward controller can predict the disturbance before it reaches the passenger and generate a cancelling signal. If only local microphone signals are available, feedback or hybrid feedforward–feedback control may be required.

3.0.8 Implications for the InnovAero VIP cabin concept

The reviewed literature leads to several concrete implications for the VIP-cabin design problem.

First, the baseline acoustic model should distinguish between primary noise propagation and secondary control propagation. The team should explicitly define $P(z)$ for the dominant engine-to-ear path and $S(z)$ for the actuator-to-ear path. Even if the numerical calculation is simplified, this notation makes the modelling approach consistent with ANC literature.

Second, passive and active treatments should be assigned to different frequency ranges. Passive insulation and absorbing panels should be used for broadband mid/high-frequency components, while active control should target low-frequency tonal or narrowband engine-related components. This frequency split is technically credible and consistent with the established ANC literature.

Third, local quiet-zone control is more realistic than global cabin control. A VIP compartment is smaller than a conventional economy cabin section, but the passenger’s ear locations remain the primary comfort targets. A seat-integrated or headrest-integrated system can reduce complexity and hardware weight while directly improving perceived comfort.

Fourth, the reclined sleeping position should be considered as a second acoustic target. If the head position changes from seated to reclined, the transfer paths $P(z)$ and $S(z)$ change as

well. The design should therefore include either multiple error targets, virtual sensing, or a mode switch between seated and reclined positions.

Fifth, active trim panels or sidewall modules are attractive if the dominant sound path enters through the window-side fuselage or cabin lining. Such modules may be more integrated and less visible than additional loudspeakers, but they require careful actuator placement and structural-acoustic modelling.

Finally, modern machine-learning-assisted ANC is promising but should be presented carefully. Deep learning or classifier-based filter selection can be framed as an optional enhancement for changing flight conditions, not as the sole basis of the solution. For an engineering competition, a robust classical FxLMS or frequency-domain FxLMS formulation should remain the mathematical core, while machine learning can be introduced as an innovation layer.

3.0.9 Summary of reviewed literature

Table 3.2: Summary of the reviewed literature and relevance to the VIP cabin task.

Reference	Main contribution	Relevance to VIP cabin concept
Kuo and Morgan [5]	Tutorial review of ANC, adaptive filters, and FxLMS	Mathematical foundation for feedforward ANC and secondary-path compensation
Elliott and Nelson [4]	Physical and signal-processing limits of ANC	Justifies active control for low-frequency cabin noise
Nelson and Elliott [6]	Theory of active sound control in physical fields	Explains quiet-zone limitations and spatial-control requirements
Johansson [7]	Aircraft-specific ANC discussion	Supports hybrid passive-active treatment in aircraft interiors
Misol [8]	Aircraft interior ANC using frequency-domain FxLMS	Guides sensor/actuator placement and low-frequency tonal control
Kletschkowski [9]	Overview of low-frequency active control in aircraft cabins	Supports low-frequency focus for engine and flow-related cabin noise
Dimino et al. [10]	Seat-headrest ANC with two loudspeakers and 2x2 FxLMS	Directly applicable to VIP seat/headrest quiet-zone design
Mylonas et al. [11]	Low-complexity local ANC for adjacent aircraft seats	Supports computationally efficient modular seat-based ANC
Mylonas et al. [12]	Virtual sensing for aircraft-seat headrest ANC	Enables ear-level control without intrusive microphones
Misol and Algermissen [13]	Active sidewall panels and virtual microphones	Supports lining-integrated active modules
Wang et al. [14]	Deep-learning-assisted active trim panels	Innovation option for multi-tonal, time-varying engine noise
Liu et al. [3]	Structural-intensity method for passive/active box-structure control	Helps identify dominant cabin panel paths and treatment locations
Parker Aerospace [2]	Industrial discussion of passive, active, and hybrid aircraft noise solutions	Engineering justification for combining passive materials and active low-frequency control

Overall, the literature indicates that the most convincing solution for the VIP cabin is a hybrid architecture. Passive sound-absorbing and sound-insulating materials should enclose the compartment to reduce broadband sound transmission and improve baseline comfort. Active noise control should then be used locally, preferably around the passenger’s head in the seated and reclined positions, to suppress low-frequency tonal or narrowband noise that is inefficient to attenuate passively. The mathematical core should be a feedforward or hybrid ANC controller

based on FxLMS or frequency-domain FxLMS, extended if necessary by MIMO control, virtual sensing, and flight-condition-dependent filter selection. This approach directly connects the acoustic cabin problem with adaptive digital signal processing and digital communication theory.

4 System Concept

4.1 Concept Overview

The proposed concept is a hybrid acoustic system for a VIP cabin zone. Passive interior elements provide broadband attenuation and structural damping, while a localized active control layer addresses the remaining low-frequency disturbance. This division of labor keeps the active system focused on the portion of the spectrum where it provides the highest value.

4.2 Architecture

A practical system can be organized into four blocks.

1. A reference path that captures or predicts the incoming disturbance.
2. A controller that computes the anti-noise signal.
3. Secondary-path actuators and loudspeakers placed near the passenger zone.
4. Error sensing or virtual sensing to evaluate the local residual sound field.

4.3 Why This Architecture Fits the Task

This architecture is suitable for the competition because it is modular, scalable, and easy to explain in an academic report. It can be deployed around a specific VIP seating area instead of the full aircraft interior, which reduces hardware count and integration effort. It also provides a natural bridge between classical ANC literature and a premium-cabin design context.

4.4 Design Principles

The concept should obey the following principles.

- Keep the control zone local and well-defined.
- Use passive treatment to reduce the required active effort.
- Favor low-order, computationally efficient adaptive control.
- Design for maintainability, not only for peak laboratory performance.

5 Methodology

5.1 Analysis Approach

The methodology should combine literature review, acoustic reasoning, and a validation-oriented design workflow. The first step is to identify the dominant cabin noise components and the likely transmission paths. The second step is to select a control topology that is compatible with the cabin layout. The third step is to estimate the benefit, complexity, and risk of the proposed design.

5.2 Evaluation Metrics

A professional assessment should use metrics that are meaningful for both engineering and passenger comfort.

- Residual sound pressure level in the target zone.
- Frequency-selective attenuation in the low-frequency band.
- Computational cost of the control update.
- Sensitivity to modelling errors and sensor placement.
- Integration impact on mass, volume, and maintenance effort.

5.3 Validation Plan

A clear validation plan can be built in three levels. First, a simplified analytical model is used to explain the control principle. Second, a simulation model evaluates the system under representative disturbance scenarios. Third, a hardware-oriented discussion checks whether the selected architecture can be integrated into a real cabin environment.

5.4 Expected Report Outputs

The final competition report should include the assumption set, the control block diagram, the evaluation criteria, and the rationale for choosing the final architecture. If data becomes available later, the same structure can be extended with simulation plots, frequency response results, and a comparison against alternative designs.

6 Conclusion and Outlook

6.1 Conclusion

The literature indicates that no single technique is sufficient for a premium aircraft cabin. Passive measures are robust, active control is effective in the low-frequency range, and hybrid systems provide the best balance between performance and feasibility. For an InnovAero submission, the strongest position is therefore a localized hybrid concept supported by a transparent methodology.

6.2 Outlook

The next development step should be a more detailed concept study with explicit geometry, control-zone definition, and disturbance assumptions. After that, the report can be expanded with simulation results, a comparison of candidate architectures, and a refined implementation discussion for Lufthansa Technik review.

Bibliography

- [1] Lufthansa Technik, “Innovaero competition 2026: Task description,” 2026, competition task document provided by the organiser.
- [2] Parker Aerospace, “Passive, active and hybrid solutions for aircraft interior noise problems,” White Paper, n.d., accessed for related-work preparation. [Online]. Available: https://www.parker.com/content/dam/Parker-com/Literature/noise-vibration---harshness-division/aerospace---defense/white-papers---case-studies/White-Paper---Passive-Active-and-Hybrid-Solutions-for-Aircraft-Interior-Noise-Problems_LL6007.pdf
- [3] Z. S. Liu, H. P. Lee, and C. Lu, “Passive and active interior noise control of box structures using the structural intensity method,” *Applied Acoustics*, vol. 67, no. 11–12, pp. 1121–1141, 2006.
- [4] S. J. Elliott and P. A. Nelson, “Active noise control,” *IEEE Signal Processing Magazine*, vol. 10, no. 4, pp. 12–35, 1993.
- [5] S. M. Kuo and D. R. Morgan, “Active noise control: A tutorial review,” *Proceedings of the IEEE*, vol. 87, no. 6, pp. 943–973, 1999.
- [6] P. A. Nelson and S. J. Elliott, *Active Control of Sound*. London: Academic Press, 1992.
- [7] S. Johansson, “Active noise control in aircraft,” Ph.D. dissertation, Blekinge Institute of Technology, 2015. [Online]. Available: <https://www.diva-portal.org/smash/get/diva2:838222/FULLTEXT01.pdf>
- [8] M. Misol, “Design of an active noise control system for aircraft interior noise,” in *Proceedings of the 29th International Congress on Sound and Vibration (ICSV29)*, 2023. [Online]. Available: <https://elib.dlr.de/195890/>
- [9] T. Kletschkowski, “Some insights in active control of noise and vibration in aircraft cabins,” in *Proceedings / Repository entry, Hamburg University of Applied Sciences*, 2020. [Online]. Available: <https://reposit.haw-hamburg.de/handle/20.500.12738/10681>
- [10] I. Dimino *et al.*, “Active noise control for aircraft cabin seats,” *Applied Sciences*, vol. 12, no. 11, p. 5610, 2022.
- [11] D. Mylonas *et al.*, “A low computational complexity local active noise control system for adjacent aircraft seats,” in *Forum Acusticum 2023*, 2023. [Online]. Available: <https://dael.euracoustics.org/confs/fa2023/data/articles/000022.pdf>
- [12] —, “A virtual sensing active noise control system based on a local headrest application,” *Journal of Vibration Engineering & Technologies*, 2024.
- [13] M. Misol and S. Algermissen, “Active sidewall panels with virtual microphones for aircraft interior noise reduction,” in *Conference paper / aircraft interior noise reduction*, 2020. [Online]. Available: https://www.researchgate.net/publication/344634967_Active_Sidewall_Panels_with_Virtual_Microphones_for_Aircraft_Interior_Noise_Reduction
- [14] B. Wang *et al.*, “Deep learning-based active trim panels for aircraft noise control,” in *AIAA Aviation Forum 2025*, 2025. [Online]. Available: <https://elib.dlr.de/215696/>